CURRICULUM VITAE · VOL. 2

Mark Akritas

data scientist · ml & ai engineer

fitting models to a noisy universe.

now ai/ml engineer @ advance2000
focus llms · rag · agents · voice ai
codes python · r · sql · js
speaks hy · en · ru · el
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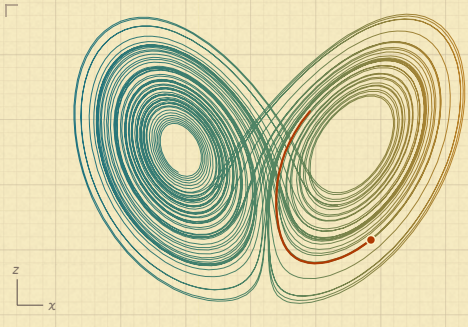


fig. 1 — *the lorenz attractor, σ=10 ρ=28 β=8/3. two particles started 10^{-6} apart part ways within twenty turns. in the interactive edition you can drag it.*

§0 abstract

Data scientist and ML/AI engineer. Working with data since 2019: experiment design and mixed-effects models, ML pipelines and signal processing, now LLMs: fine-tuning, RAG, agents and voice AI. Master's in industrial engineering; before that, crisis management and accounting. Interests: music, history, science & philosophy.

KEYWORDS llms · rag · agents · voice ai · lora · mixed-effects models · design of experiments · signal processing · python · r

§1 work experience

chords: skill families in play

2025.11 — now
1 yr

AI/ML Engineer

Advance2000 · cloud private infrastructure

LLM fine-tuning using LoRA. A private, ChatGPT-like AI assistant with web search, RAG and agentic features. Voice AI agents for call centers. Codebase and task management across the team for multiple projects.

Python · LoRA · PyTorch · LLMs · vLLM · RAG · Weaviate · SearXNG · agents · LiveKit · ElevenLabs · Deepgram · Git · Docker · Linux · Pydantic · uv · ruff · VS Code

2025.01 — 2025.11
10 mos

AI Engineer

NewLevelWork · ai coaching, leadership & development

LLM- and voice-based chatbots, agentic workflows, RAG, document parsing, REST APIs.

Python · LLMs · agents · RAG · kreuzberg · Weaviate · LiveKit · ElevenLabs · Deepgram · REST · Docker · Git · Pydantic · uv · VS Code

2020.06 — 2026.05
5 yrs 11 mos

Data Scientist / ML Engineer

NobleScripts · ai & web dev

Research, machine learning, signal processing, REST API development, data collection, pipelines, deployment, codebase & task management, long-term maintenance.

Python · SQL · JS · Shell · ML · sklearn · PyTorch · signals · REST · data eng. · Docker · Linux · Git · Jupyter · VS Code · uv · ruff

2019.06 — 2020.06
1 yr

Data Scientist / Statistician

ClinStatDevice / ClinSoft · bioinformatics

Design of experiments, randomization, simulation, mixed-effects modelling, advanced visualization, dashboard development, report automation.

R · SQL · DoE · mixed models · dashboards

§1′ other experience

IT Intern · Veon Armenia
2018.09 — 2018.11 · 2 mos
SQL reports, natural language processing.

Teaching Associate · AUA
2021.01 — 2021.05 · 4 mos
Programming for Data Science: teaching Python & R.

Research Assistant · PDRF
2020.07 — 2021.03 · 8 mos
Skill-mismatch research: survey preparation, data analysis.

EPIC Startup Program · MetaStyle
2020.01 — 2020.06 · 5 mos
Text style transfer using machine learning.

AI Incubator Program · Hero House · Expper
2019.07 — 2019.10 · 3 mos
Robin the Robot: human-robot interaction research, emotional-response system prototyping.

Digitalization of Accounting · Wikimedia Armenia
2017.06 — 2017.08 · 2 mos

Accountant · IT, retail & public
2015.08 — 2017.09 · 2 yrs 1 mo

Judicial Acts Specialist
2014.03 — 2015.08 · 1 yr 5 mos

Military Service · Armed Forces of RA
2011.12 — 2013.12 · 2 yrs

continued overleaf: the career as a score, and the stack as a sky →

§1 trajectory

fig. 2

role	where	dates
INDUSTRY		
AI/ML Engineer	Advance2000	2025.11–now
AI Engineer	NewLevelWork	2025.01–2025.11
Data Scientist / ML Engineer	NobleScripts	2020.06–2026.05
Data Scientist / Statistician	ClinStatDevice / ClinSoft	2019.06–2020.06
IT Intern · SQL, NLP	Veon Armenia	2018.09–2018.11
RESEARCH & TEACHING		
Teaching Associate	AUA	2021.01–2021.05
Research Assistant	PDRF	2020.07–2021.03
EPIC Startup Program	MetaStyle	2020.01–2020.06
AI Incubator	Hero House · Exppper	2019.07–2019.10
EDUCATION		
Master, IE & Systems Mgmt	AUA	2017.09–2020.06
Bachelor, Crisis Management	ASUE	2010.09–2016.06
BEFORE DATA		
Digitalization of Accounting	Wikimedia Armenia	2017.06–2017.08
Accountant	IT, retail & public	2015.08–2017.09
Judicial Acts Specialist		2014.03–2015.08
Military Service	Armed Forces of RA	2011.12–2013.12
pooled estimate	data & ml	2019.06–now · 7 yrs 5 mos

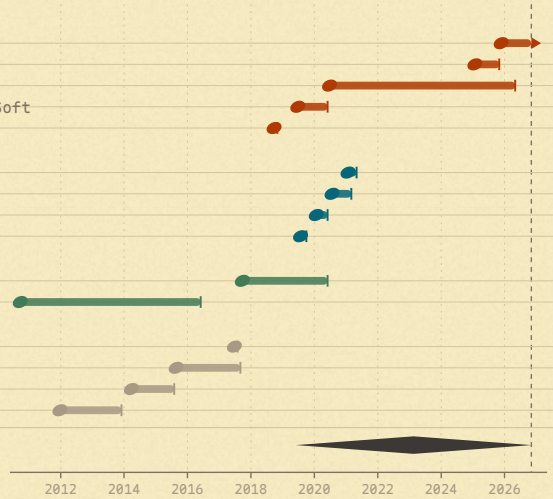


fig. 2 — *career as a piano roll: rows are roles, notes span tenure, chords count skill families; ◆ pools the data years.*

§2 methods

fig. 3

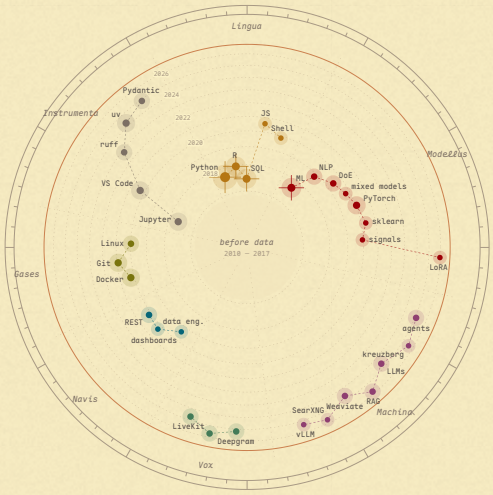


fig. 3 — *the stack as a sky: angle = family, radius = first year in use, size = roles using it.*

THE STACK, BY CONSTELLATION

- ***Lingua*** · languages
  Python, R, SQL, JavaScript · HTML/CSS, Shell
- ***Modellus*** · modelling
  machine learning, NLP, design of experiments, mixed-effects models, PyTorch, scikit-learn, signal processing, LoRA fine-tuning
- ***Machina Loquens*** · llm stack
  agents, kreuzberg, LLMs, RAG, Weaviate, SearXNG, vLLM
- ***Vox*** · voice ai
  Deepgram, ElevenLabs, LiveKit
- ***Navis*** · shipping
  dashboards, data engineering, REST APIs
- ***Gases Nobiles*** · noble gases
  Docker, Git, Linux
- ***Instrumenta*** · tooling
  Jupyter, VS Code, ruff, uv, Pydantic

PROGRAMMING · SELF-ASSESSED

Python — *excellent*	R — *good*
SQL — *intermediate*	JavaScript · HTML/CSS — *basic*

§4 education

2017.09 – 2020.06
Master in Industrial Engineering & Systems Management
American University of Armenia
CONCENTRATION data analytics
CAPSTONE temporal data-based anomaly localization in networks

2010.09 – 2016.06
Bachelor in Crisis Management
Armenian State University of Economics
THESIS spatial development in the Republic of Armenia

§5 languages

fig. 5

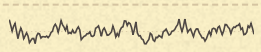
Հայերեն armenian		native or bilingual *level 5/5*
Русский russian		native or bilingual *level 5/5*
English english		full professional *level 4/5*
Ελληνικά greek		limited working *level 2/5*

fig. 5 — *amplitude ~ proficiency (0–5), noise ~ the effort it still takes.*

§3 projects

NewsPEPper
weekly python newsletter
PEPs and Python releases, summarized every saturday.
newspepper.dev ↗

map of philosophy
in development for philosophy enthusiasts
Who read whom, from heraclitus to wittgenstein.

jumpwise
vs code extension
Keyboard navigation between the symbols of a file.
vs code marketplace ↗

§6 marginalia

INTERESTS
history · music · philosophy · science · linux ricing

set in maple mono · gruvbox on paper. every figure is live in the interactive edition:
markakritas.github.io